



Purchasing: Freight

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

Welcome to the topic on managing freight in SAP Business One.

Objectives



At the end of this topic, you will be able to:

- Describe how freight charges are managed in business processes
- Set up freight
- Add freight in sales and purchasing documents

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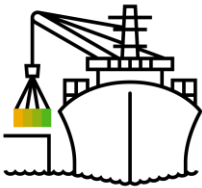
Business Scenario



- Your company buys and sells items throughout the country
- When items are shipped to your warehouse, you get charged for the shipping and delivery. You want to record these charges, as well to have the inventory unit cost reflect the additional charge.
- Some items that you sell are high value and require insurance during shipping. You charge your customers a fixed amount for insurance, however you do not want the insurance cost to affect unit cost.

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Freight definition



Freight is a charge paid for carriage or transportation of goods by air, land, or sea.

You can also use freight to apply any additional expenses that are not part of the basic item price such as insurance of the goods.

Freight can be added on any purchasing or sales document.

Total Before Discount	GBP 800.00
Discount %	
☒ Total Down Payment	
Freight	GBP 245.00
☐ Rounding	GBP 0.00
Tax	
Total Payment Due	GBP 1,045.00
Applied Amount	
Balance Due	GBP 1,045.00

Freight is a charge paid for carriage or transportation of goods by air, land, or sea. You can also use freight to apply any additional expenses that are not part of the basic item price such as insurance of the goods.

Freight can be added to any of the sales and purchasing documents.

Freight Charges in Business Processes

Freight can be added to the total document or on the row level.

Special handling item 1	50
Special handling item 2	25
General shipping	125
Insurance	45
Total freight	245

The screenshot displays the SAP Freight Charges interface. The top section shows a table with columns for Item No., Quantity, Unit Price, Dis., Tax, Total (LC), Bin Locati., and Summary Type. The bottom section shows a summary table with columns for Total Before Discount, Discount, Freight, Tax, and Total Payment Due. An orange circle highlights the 'Freight' row in the summary table, and an orange arrow points from this row to the 'Freight' row in the item-level table above.

Item No.	Quantity	Unit Price	Dis.	Tax	Total (LC)	Bin Locati.	Summary Type
1	5	GBP 100.00	0.00	II	GBP 500.00	5	Freight 1
2	2	GBP 150.00	0.00	II	GBP 300.00	2	Freight 1 (LC)
3			0.00	II			

#	Freight	Remarks	Tax G.	Tax...	Total Tax A...	Distrib. M...	Net Amount	Sta...	Gross Am...	De...
1	Freight	General	II	20.00	GBP 25.00	Row Tot...	GBP 125.00	O	GBP 150.00	
2	Insurance		II	20.00	GBP 9.00	Row Tot...	GBP 45.00	O	GBP 54.00	
							GBP 170.00		GBP 204.00	

	Amount
Total Before Discount	GBP 800.00
Discount	
Freight	GBP 75.00
Tax	GBP 0.00
Total Payment Due	GBP 1,050.00

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Freight can be added to the total document or on the row level for specific items.

You can apply freight in both purchasing and sales documents. You may charge freight when shipping items from your warehouse or your vendor may add freight charges to items you receive.

For example, we see an A/P invoice with freight added. Each item had a special handling charge in addition to the overall shipping charge that was added at the document level. The special handling items were added on the rows. The overall charges for freight and insurance were added by choosing the link arrow on total freight. The amount on the total document and the rows combine to give the total for freight in the document.

Accounting and Inventory Effects

Different types of freight can be defined to post to different accounts.

For example:

- Insurance might post to an expense account.
- Shipping costs might be set up to post to the stock account for the item.
In this way, the freight cost becomes part of the unit cost of the item.

#	G/L Acct/BP ...	G/L Acct/BP Name	Debit	Credit
1	→ V10000	Acme Associates		GBP 1,254.00
2	→ 143030	VAT Receivable (Input Tax)	GBP 209.00	
3	→ 130000	Raw Materials	GBP 1,000.00	
4	→ 600000	Inbound Freight & Carriage	GBP 45.00	
			GBP 1,254.00	GBP 1,254.00

Cost of item plus
shipping cost
posting to stock
account

Insurance cost
posting to freight
expense account

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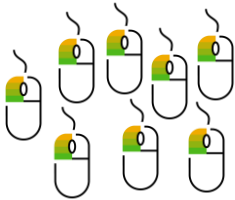
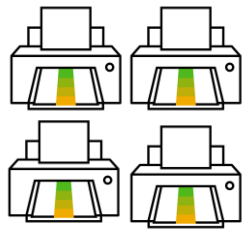
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You have the option to define different types of freight charges. One type of freight might post to an expense account while another type could be set up to post to the stock account for the item. In the latter, the freight cost becomes part of the unit cost of the item.

In our previous example, freight shipping charges were set up to be posted to the item's stock account along with the cost of the item. Insurance charges were set up to be charged to an expense account.

Both the cost of the items (800) and the freight shipping charges (200) post to the item stock account. Insurance is posted instead to an expense account that has been specified in the setup window.

Incorporating freight into item cost

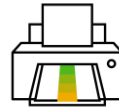


You can distribute total freight to rows and incorporate the costs into the item's unit cost.



Example:

Document shipping cost: 100
Shipping costs applied to items based on row weight



Row 1 is 80% of weight in total shipment

Cost of 80 is posted to the stock account for the item.



Row 2 is 20% of weight in total shipment

Cost of 20 is posted to the stock account for the item

You may wish to incorporate the freight costs into the cost of the item. It is easy to factor in the freight cost if it is on the row level. However, when freight is charged at the document level, we need a method for how to distribute those costs to the individual items. The freight amount added to each line item in the document is calculated based on a distribution method that you define while setting up freight.

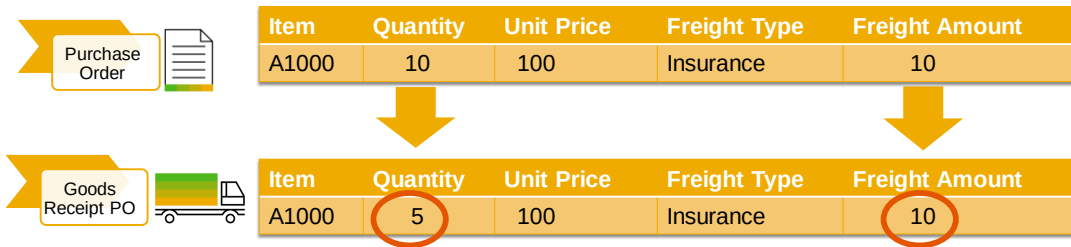
For example, OEC Computers regularly buys electronic equipment to resell. Freight costs are factored into each item's cost. This ensures that all freight costs related to the item are taken into account when viewing the cost of goods sold.

When multiple items are purchased on the same document, the total freight costs needs to be distributed among the rows. Most freight costs charged to OEC are based on weight. Therefore, OEC uses a distribution method for freight based on the weight of the items on the rows.

Copying Freight Charges

Based on the freight definition, row freight charges are copied fully on the first reference, partially or not at all to subsequent documents.

In our example, the freight type insurance definition is set to copy the full amount the first time that row is copied even when the row quantity is reduced.



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In our example, we saw freight added in an A/P invoice. Freight might be added at an earlier point, for example in a Goods Receipt PO based on a shipping document.

On the sales side, we might charge freight at the point of sale. In that case, the sales order would contain the freight charges that would be then copied over to the delivery and A/R invoice.

When you have freight charges entered on the row level and you use the *Draw Document Wizard* to copy from one document to another, the row level freight charges will be copied to the target document based on the freight type's definition. You have the option to define a freight type to either draw the amount fully the first time a row is partially copied, partially (in relation to the total or the quantity copied) or not at all.



Freight Setup



Now let's see how to set up freight in SAP Business One

Activation in Document Settings

Administration → System Initialization → Document Settings
On the General Tab, choose Manage Freight in Documents



The screenshot shows a user interface for document settings. At the top left, the word 'General' is displayed. Below it, there is a checkbox with a checkmark inside, followed by the text 'Manage Freight in Documents'. To the right of this checkbox is a grey button labeled 'Freight - Setup'.

You cannot reverse this setting after a document has been posted.

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To start using freight, you first must activate it in *Document Settings*. This setting will add a *Freight* field in all purchasing and sales documents, and allow you to calculate additional costs connected to a document. Once you have selected this checkbox you cannot reverse it after a document has been posted.

Freight Setup

Use the button **Freight - Setup** on the **General Tab** of **Document Settings**

Or use the menu path: **Administration → Setup → General → Freight**

Name	Revenue and Expense accounts / Tax Groups	Fixed Amount Revenue / Expenses	Gross Freight	WTax Liabie	Distribution Method	Drawing Methods	Stock / Last Purchase Price checkboxes
Shipping	Revenue/Expense account		√		Volume	Quantity	√
Insurance	Revenue/Expense account	5			None	Total	

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In the *Freight Setup* window, you need to create a row for each type of freight charge you plan to track. In our business example, we would need to set up two rows, one for shipping charges and the second for insurance that is used to insure high value items during transport.

After you enter the name, select an expense account to be used in purchasing documents and a revenue account to be used in sales documents.

You also select a default tax group for incoming and outgoing documents, to be used if one does not default in from the *Business Partner Master Data*.

You can define a fixed expense amount that is added automatically to every sales or purchasing document. You can modify this amount in the document if needed.

Next you can choose whether the fixed amount you entered includes tax, by checking the gross freight box.

You can also mark whether the costs are subject to withholding tax.

Then choose one of the six distribution methods. A distribution method is used when freight is entered in the document header and is set up to affect stock. We will see an example in the upcoming slide.

In the *Drawing Method*, specify the required portion of the freight to apply when you partially copy a base document to a target document. We will see more details in upcoming slides.

Select the *Stock* checkbox in order for freight charges to be added to the item's unit cost, when the item is purchased. This is applicable for items with a valuation method of FIFO, moving average or serial and batch valuation. If freight charges are added to a standard cost item, the amount will be recorded as an expense.

You can check the *Last Purchase Price* checkbox if you wish the freight charges to be added to the items' last purchase price in a purchasing document.

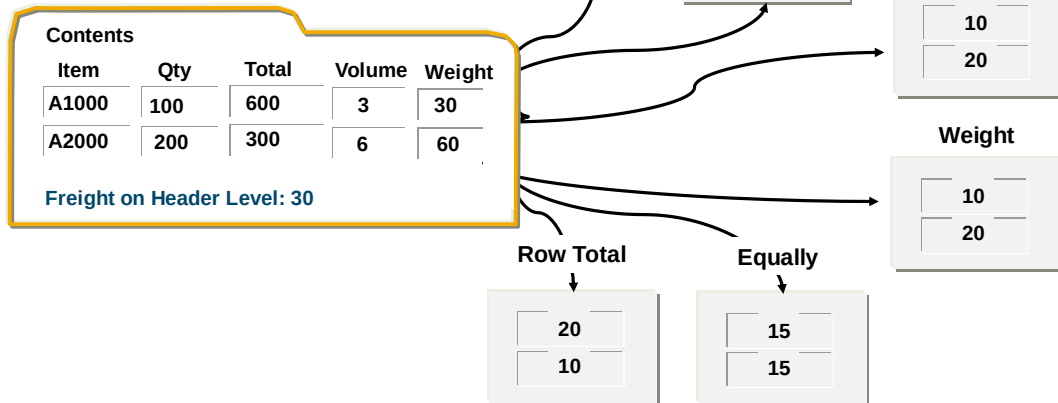
If the either the *Stock/Fixed Asset* or *Last Purchase Price* checkboxes are selected, a distribution method (other than None) must be selected.

You can optionally assign a distribution rule, or a project code, to be assigned to the freight revenue or expenses account row in the journal entry. Note, freight charges that affect stock, such as shipping in our example, used in a document with inventory item, will not appear as a separate row in the journal entry, therefore no distribution rule or project code will be assigned.

Based on our business scenario, we will define two (2) freight charges, the first one will be for shipping, without a fixed amount, and it will affect stock. The second for insurance cost that you charge your customers during the sales cycle. Here we enter a fixed amount, since most deliveries have insurance.

Freight setup – Distribution Method

Examples of Distributed Amounts
for each Distribution Method



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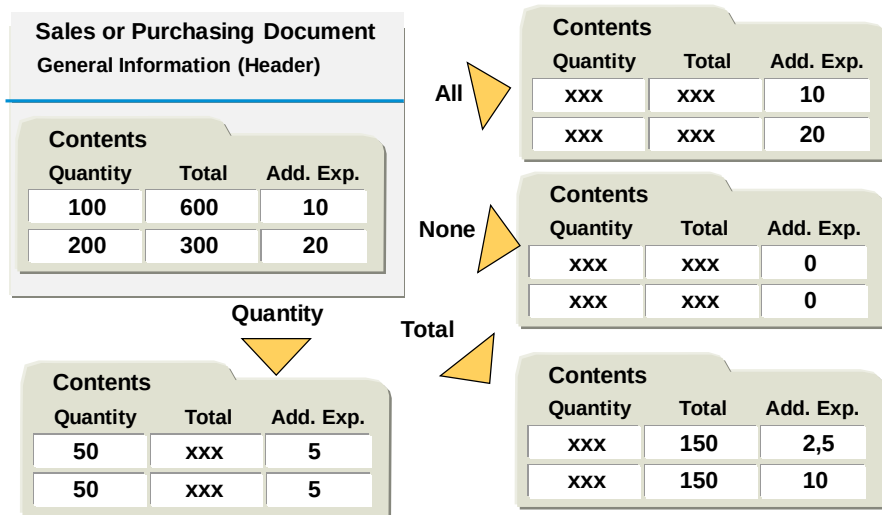
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Let's look at how freight is distributed between the rows in a purchasing document for each of the 6 distribution methods.

In this example, the amount of 30 is entered as freight on the document header.

- None - Freight expenses are not distributed to the line items.
- Quantity - Expenses are distributed to the line items according to the relation of the quantities per line item.
- Volume - Expenses are distributed to the line items according to the relation of the volumes per line item.
- Weight - Expenses are distributed to the line items according to the relation of the weights per line item.
- Equally - Expenses are distributed to the line items equally.
- Row Total - Expenses are distributed to the line items according to the relation of the row totals of the line items.

Freight Setup – Drawing Method



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Let's look at the four options for Drawing Methods.

When you use the *Draw Document Wizard* to copy from one document to another, freight entered in the row level will be copied to the target document based on the proportionate quantity that was drawn.

The drawing method determines how the system copies the freight expenses or revenues from the base document to the target document in case you only partially copy the row items. In the graphic, we see the results based on each method.

- **All** - The system copies the entire amount the first time you copy any partial amount of the line item.
- **None** – When the drawing method is None, nothing is taken into the target document.
- **Total** - If you copy a line item only partially, the system copies the proportionate amount based the **total** you copied. When you copy the outstanding line items, the system copies the remaining amount.
- **Quantity** - If you only partially copy a line item, the system copies the proportionate amount based on the **quantity** you copied. When you copy the outstanding line items, the system copies the remaining amount.



Adding Freight in Marketing Documents



Now that we've seen how the setup works, let's review the process of adding freight in marketing documents.

Adding Freight in Sales Documents - Rows

Row level freight in Sales

Contents		
Item	Quantity	Total
A1000	10	100
A2000	5	50

Freight 1	Freight 1 LC	Freight 2	Freight 2 LC	Freight 3	Freight 3LC
Freight	10	Insurance	5	Other	8
Freight	5	Insurance	2	Other	8

Total Freight (From Row) $10+5+8+5+2+8 = 38$

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Since there are a few differences between sales and purchasing documents, let's review each type separately.

Let's look at what happens when we enter freight in the row level of sales documents.

You have the ability to define up to 3 kinds of freight in the row level. The columns for the freight and freight amount can be added using *Form Settings*.

First you choose the freight charges using a dropdown, and then you enter the freight amount for the particular row.

You can add additional information regarding each freight charge such as a freight distribution rule, a project code, a tax code, and display a tax amount.

The total freight entered in the rows is displayed in the *Freight* field, at the bottom of the document. The total includes row and additional header freight amounts.

Adding Freight in Sales Documents - Header

Header level freight in Sales Documents:

Open the *Freight Charges* by clicking the drill down arrow next to the Freight field

Freight Name	Remarks	Tax Code/ Amount	Distribution Method	Net Amount	Status	DL/Project
Freight	Expedited delivery	Exempt	Volume	5	O	
Insurance	High Value Items	Exempt	None	5	O	

Set the *Distribute Freight* to Yes on the row to distribute the header freight to that row

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To view and maintain freight charges for the document header, use the drill down arrow next to the *Freight* field.

In this window, all the freight types that you defined during the setup are displayed.

If you entered a fixed amount for revenues during the setup, it will be added to all sales documents. You can delete the amount or add to that amount as needed in the Net Amount column.

The distribution method is also automatically populated based on the freight setup, and can be modified here.

Note in a sales document, the purpose of distributing freight amounts between the rows is informative only.

In sales document's rows, you can find a column called *Distribute Freight*. If set to Yes, the freight amount in the header is distributed to the row, based on the distribution method defined here. In fact the field's default is set to No, however if it is important for you to see the impact of freight in the row level, change this field to Yes.

You can view the status of the freight line here as well. Note that it's possible to have a closed freight line in an open document.

Lastly if a distribution rule or freight was defined during setup, it will appear here, and can also be modified in documents. In the journal entry, the *Freight Revenue* account shows as a separate row and it will be assigned with a distribution rule or a project code, if one was assigned here.

The total freight amount displayed in the document includes the row and the header freight charges, however when you drill down to the freight charges window, only the header freight details display.

Adding Freight in Purchasing Documents - Rows

Row level freight in purchasing:

Contents

Item	Qty	Total	Freight 1	Freight 1 LC	Last Purchase Price	Inventory
A1000	10	100	Freight	10	Yes	Yes
A2000	5	50	Freight	5	No	No

In purchasing documents, there are additional two columns related to freight that do not exist in sales documents: *Last Purchase Price* and *Inventory*.

The values (yes or no) are copied from the freight setup. If you wish to override the default value, for example for one particular purchase, you can add these columns, using *Form Settings*, and change the default for each row.

Other than these 2 columns, the rest of freight fields are the same as in sales documents.

Adding Freight in Purchasing Documents - Header

Header level freight in Purchasing:

Open the *Freight Charges* by clicking the drill down arrow next to the *Freight* field

Freight Name	Remarks	Tax Code/ Amount	Distribution Method	Net Amount	Status	DL/Project
Freight	Expedited delivery	Exempt	Volume	5	O	
Insurance	High Value Items	Exempt	None	5	O	

Freight in the purchasing document header is also similar to freight in the sales document header.

If the freight is not set up to affect stock or last purchase price, such as in our scenario for the insurance, the freight amounts will be recorded as expenses.

Freight Charges Journal Entry

A/R Invoice (based on a Delivery):

Account	Debit	Credit	Project/Distribution Rule
Customer	110		
Revenue		100	
Freight Revenue 1		5	Project Code/Distribution Rule
Freight Revenue 2		5	Project Code/Distribution Rule

A/P Invoice (based on a Goods Receipt PO):

Account	Debit	Credit	Project/Distribution Rule
Vendor		110	
Inventory	105		
Freight Expense	5		Project Code/Distribution Rule

Let's take a look at the journal entries that are created on sales and purchasing documents when adding freight.

On the sales side, if we assume our freight amount is 5 for shipping and 5 for insurance, the freight amounts are posted on the credit side to the freight revenue account as defined in the freight setup. If a project code and a distribution rule were selected, this displays in the journal entry row for the freight revenues.

On the purchasing side, in our example we have a freight amount of 5 that affects inventory, and 5 that doesn't, therefore the first 5 will be added to the inventory account, and the second amount of 5 will be posted on the debit side to the freight expense account.

If a project code and a distribution rule were selected, these will only apply to the freight expense that did not affect inventory.

Summary



Here are some key points to take away:

- Freight is used to record additional charges on sales and purchasing documents
- Freight can be added to the document row or header
- Header level freight can be allocated to the rows as needed using different allocation methods
- Freight can affect inventory item's unit cost and last purchase price

Here are some key points to take away:

- Freight is used to record additional charges on sales and purchasing documents
- Freight can be added to the document row or header
- Header level freight can be allocated to the rows as needed using different allocation methods
- Freight can affect inventory item's unit cost and last purchase price



Test your knowledge



Test your knowledge – Freight

Freight can be in a sales document...

1. In the document header only
2. In the row level only
3. In document header and row level
4. In document header and inventory items row only

Test your knowledge – Freight

Freight can be in a sales document...

1. In the document header only
2. In the row level only
3. In document header and row level
4. In document header and inventory items row only

Test your knowledge – Freight

When you enter freight for the row, no tax code or group are displayed. Why?

1. You didn't check the Gross Freight in the freight setup
2. You checked the Gross Freight in the freight setup
3. In a purchasing document, tax is not calculated on freight
4. The tax code or groups were not marked to affect freight

Test your knowledge – Freight

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1. You didn't check the Gross Freight in the freight setup
2. You checked the Gross Freight in the freight setup
3. In a purchasing document, tax is not calculated on freight
4. The tax code or groups were not marked to affect freight

Test your knowledge – Freight

True/False – you can only assign freight charges with a project code in sales documents

1. True
2. False

Test your knowledge – Freight

True/False – you can only assign freight charges with a project code in sales documents

1. True
2. False

Test your knowledge – Freight

True/False – row level freight is copied in full when the document is converted to a higher level document

1. True
2. False

Test your knowledge – Freight

True/False – row level freight is copied in full when the document is converted to a higher level document

1. True
2. False

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